Math 571 - Homework 2

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Problems numbered [B:2.1:3] are from Bergman's exercise notes. Problems labeled [R:2.1] are directly from Rudin. If I add an asterisk (*) to a problem, we have a slight variant of the problem.

Problem 2.1 (R:2.2*)**.** A complex number γ is **algebraic** iff γ is a root to a polynomial with integer coefficients. Prove that there are complex numbers that are not algebraic.

The following are two fun facts just beyond what we got to in MAT-513:

Let $\mathbb{A} \subset \mathbb{C}$ be the set of algebraic numbers.

(a) $\mathbb{A}$ is a field.

(b) $\mathbb{A}$ is algebraically closed, that is, if α is a root of a polynomial in $\mathbb{A}[x]$, then $\alpha \in \mathbb{A}$. So in the definition of algebraic numbers you can use any ring of coefficients R, $\mathbb{Z} \subseteq R \subseteq \mathbb{A}$.

This can be put together as a sequence of steps with results in the text.

There are only countably many polynomials with integer coefficients. **Why? Justify this.**

Each of the countably many polynomials has at most finitely many roots. **Why? Here, you can just state this, but you should reflect on how you know this. This is a MAT-513 fact.**

Putting the facts together: The algebraic numbers are the countable union of finite sets, which is countable. **Why? What result gives you this?**

Problem 2.2 (B:2.2:4)**.** This problem is concerning relative topology.

(a) Suppose E is an open subset of a metric space X, and F is a subset of E. Show that F is open relative to E iff F is open.

(b) Suppose E is a closed subset of a metric space X, and F is a subset of E. Show that F is closed relative to E iff F is open.

(c) Suppose E is an open subset of a metric space X, and F is a subset of E that is closed relative to E. Show that F need not be open or closed in X.

(d) Suppose E is a closed subset of a metric space X, and F is a subset of E that is open relative to E. Show that F need not be open or closed in X.

Note: Nothing about X being a metric space is used here. This is valid for an arbitrary topological space.

(a): Recall $F \subseteq E$ is *open relative to* E iff $F = O \cap E$ for some open set in X. So if E is itself open, then $F = O \cap E$ is open. Conversely f F is open, then $F = E \cap F$ is relatively open.

(b): You can define F is relatively closed as we did relatively open. But it is more usual to use the relatively open sets to define the ***relative topology on*** E. Then F is relatively closed iff $E \setminus F$ is closed in the relative topology, that is, $E \setminus F$ is relatively open. So $E \setminus F = E \cap F^c = E \cap O$ and hence

$$F = E \setminus (E \cap O) = E \cap (E \cap O)^c = E \cap (E^c \cup O^c) = (E \cap E^c) \cup (E \cap O^c) = E \cap O^c$$

Since O^c is closed, we see that F is relatively closed iff $F = E \cap G$ for some closed G. So if E is closed, then clearly,

$$F \text{ is relatively closed} \implies F = E \cap G \text{ for a closed } G \implies F \text{ is closed}$$

Conversely, if $F \subseteq E$ is closed, then $F = E \cap F$, so F is relatively closed.

(c): Take $E = (0,1)$ and $F = (0,1/2]$. F is not open or closed in $\mathbb{R}$, but F is relatively closed in E.

(d): Take $E = [0,1]$ and $A = [0,1)$. A is not open, nor closed, but A is relatively open since $[0,1) = E \cap (-1,1)$.

Problem 2.3 (B:2.2:20)**.** Here you look at the Kuratowski game discussed in the podcast.

(a) Show that $\mathrm{Cl}(\mathrm{Cl}(E)) = \mathrm{Cl}(E)$ and $\mathrm{Int}(\mathrm{Int}(E)) = \mathrm{Int}(E)$.

(b) $\mathrm{Int}(\mathrm{Cl}(E)) = \mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(E))))$.

(c) Show that you can get at most seven different sets from applying the operations of closure and interior repeatedly to a fixed set E.

(d) Show that there is $E \subset \mathbb{R}$ so that all seven possible sets are realized.

Bonus: If you also throw in the operation of complement, then there are at most 14 possible sets you can get starting from a fixed set E.

If C is closed, then $\mathrm{Cl}(C) = C$ and if O is open, then $\mathrm{Int}(O) = O$, this is either trivial or easy depending on your definitions. The definition I would use is

$$\overline{A} = \mathrm{Cl}(A) = \bigcap \{C \mid A \subseteq C \wedge A \text{ is closed}\} = \text{the smallest closed set containing } A$$
$$A^o = \mathrm{Int}(A) = \bigcup \{O \mid O \subseteq A \wedge O \text{ is open}\} = \text{the largest open set contained in } A$$

In this case, (a) and (b) are clear.

(b) is where the action is: we have A, $\mathrm{Cl}(A)$, $\mathrm{Int}(A)$, $\mathrm{Cl}(\mathrm{Int}(A))$, $\mathrm{Int}(\mathrm{Cl}(A))$, $\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(A))$, and $\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(A)))$. Why stop here?

Claim: $\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(A)))) = \mathrm{Int}(\mathrm{Cl}(A))$ and $\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(A)))) = \mathrm{Cl}(\mathrm{Int}(A))$.

Proof of Claim: First notice that

$$(\mathrm{Int}(A))^c = \mathrm{Cl}(A^c) \text{ and } (\mathrm{Cl}(A))^c = \mathrm{Int}(A^c)$$

This is really just set algebra:

$$\begin{aligned}(\mathrm{Cl}(A))^c &= \left(\bigcap\{C \mid A \subseteq C \wedge C \text{ closed}\}\right)^c \\ &= \bigcup\{C^c \mid C \subseteq A \wedge C \text{ closed}\} \\ &= \bigcup\{C^c \mid A^c \subseteq C^c \wedge C^c \text{ open}\} \\ &= \bigcup\{O \mid A^c \subseteq C^c \wedge O \text{ open}\} = \mathrm{Int}(A^c)\end{aligned}$$

So it suffices to only prove one of the two claims. **Why? Check it!**

$\boxed{\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(\boldsymbol{A}))) \supseteq \mathrm{Int}(\mathrm{Cl}(\boldsymbol{A}))}$ Clearly, $\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(A)) \supseteq \mathrm{Int}(\mathrm{Cl}(A))$ and so $\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(A)))) \supseteq \mathrm{Int}(\mathrm{Int}(\mathrm{Cl}(A)) = \mathrm{Int}(\mathrm{Cl}(A))$.

$\boxed{\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(\boldsymbol{A}))) \subseteq \mathrm{Int}(\mathrm{Cl}(\boldsymbol{A}))}$ Start with $\mathrm{Int}(\mathrm{Cl}(A)) \subseteq \mathrm{Cl}(A)$ so $\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(A)) \subseteq \mathrm{Cl}(\mathrm{Cl}(A)) = \mathrm{Cl}(A)$. Now apply interior to both sides, $\mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(A)))) \subseteq \mathrm{Int}(\mathrm{Cl}(A))$.

This shows that there are at most seven sets obtainable from E formed by iterating the interior and closure operations.

(d) Now for our example, consider:

$$\begin{aligned}E &= [0,1) \cup (1,2] \cup ((2,3) \cap \mathbb{Q}) \cup \{4\} \\ \mathrm{Cl}(E) &= [0,3] \cup \{4\} \\ \mathrm{Int}(E) &= (0,1) \cup (1,2) \\ \mathrm{Cl}(\mathrm{Int}(E)) &= [0,2] \\ \mathrm{Int}(\mathrm{Cl}(E)) &= (0,3) \\ \mathrm{Int}(\mathrm{Cl}(\mathrm{Int}(E)) &= (0,2) \\ \mathrm{Cl}(\mathrm{Int}(\mathrm{Cl}(E))) &= [0,3]\end{aligned}$$

Problem 2.4 (B:2.2:14)**.** This investigates the p-adic metric on $\mathbb{Z}$. Let $p > 1$ be an integer. Define $d_p(s,t) = p^{-e_p(s-t)}$ where $e_p(m) = k$ iff $p^k \mid m \wedge p^{k+1} \nmid m$, so $e_p(m)$ reports the maximal number of times m can be evenly divided by p. Note that $p^k \mid 0$ for all k, so it makes sense to define $e_p(0) = \infty$ and take $d_p(s,s) = p^{-\infty} = 0$.

Show that $d_p : \mathbb{Z} \times \mathbb{Z} \to [0,\infty)$ is a metric and in fact an ***ultra-metric***, i.e., $d_p(q,s) \leq \max\{d_p(q,r), d_p(r,s)\}$.

For Fun: Show that all p-adic triangles are isosceles. This is a geometric idea; what corresponding topological fact is leading to this?

p-adics are very important and interesting number systems, particularly when p is a prime. Here is a great video introduction to p-adics, which discusses applications to solving Diophantine equations.

To check that $d_p : \mathbb{Z} \times \mathbb{Z} \to [0,\infty)$ is a metric there are three things to check:

Symmetry: That $d_p(s,t) = d_p(t,s)$ follows as $e_p(s-t) = e_p(t-s)$.

Positivity: That $d_p(s,t) > 0$ when $s \neq t$ is clear.

Triangle Inequality: Here, we argue the ultra-metric condition, which is stronger. If $p^k \mid q-r$ and $p^k \mid r-s$, then $p^k \mid q-s$ since $q-s = (q-r)+(r-s)$, so $e_p(q-s) \geq \min\{e_p(q-r), e_p(r-s)\}$ and so $p^{-e_p(q-s)} \leq \max\{p^{-e_p(q-r)}, p^{-e_p(r-s)}\}$, which is equivalent to $d_p(q,s) \leq \max\{d_p(q,r), d_p(r,s)\}$.

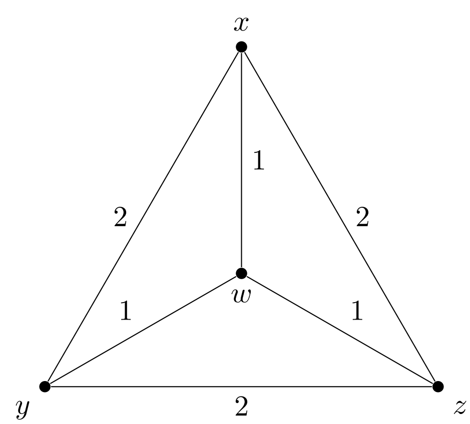
So d_p is a metric.

Extra Fun: Suppose now that q, r, s are three distinct integers. Suppose

$$d_p(q,s) = \max\{d_p(q,s), d_p(r,s), d_p(r,q)\},$$

then $d_p(q,s) \geq d_p(r,q), d_p(r,s)$. But $d_p(q,s) \leq \max\{d_p(r,q), d_p(r,s)\}$ and so $d_p(q,s) \leq d_p(r,q)$ or $d_p(q,s) \leq d_p(r,s)$. From this it is clear that $d_p(q,s) = d_p(r,s)$ or $d_p(q,s) = d_p(r,q)$. Thus the triangle formed by q, r, s is isosceles.

Problem 2.5 (B:2.2:18)**.** Let X be a 4-element set. Consider a graph on the four nodes $\{x, y, z, w\}$ with edges and edge weights as shown.



(a) Verify that $d_X : X \times X \to [0, \infty)$ is a metric where $d(x,y) = d(x,z) = d(y,z) = 2$ and $d(w,x) = d(w,y) = d(w,z) = 1$.

(b) Show that there is no $F : X \to \mathbb{R}^k$ that is distance preserving, that is, $d_X(u,v) = d_{\mathbb{R}^k}(f(u), f(v))$.

(c) The example here is a 4-point space that cannot be *isomorphically* embedded in $\mathbb{R}^k$ for any k, yet every 3-point subspace can be isomorphically embedded in $\mathbb{R}^k$. Can you find a 5-point space that cannot be isomorphically embedded in $\mathbb{R}^k$ and yet all of whose 4-point subspaces can be isomorphically embedded in $\mathbb{R}^k$ for sufficiently large k.

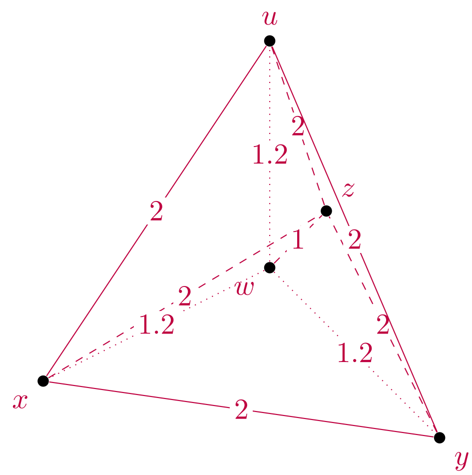
(a) Checking that d_X is easy. It is clearly symmetric and positive. There are 4 triangles to check for the triangle inequality, and it is clear that the inequality holds for each.

(b) For $d_{\mathbb{R}^k}(f(x), f(y)) = 2$ and $d_{\mathbb{R}^k}(f(x), f(w)) = 1$, then $f(w)$ must be on the $(k-1)$-sphere with center $f(x)$. Similarly, $f(w)$ is on the $(k-1)$-sphere with center $f(y)$. But these two $(k-1)$-spheres are tangent at exactly one point, namely the midpoint between $f(x)$ and $f(y)$, so $f(w) = \frac{f(x)+f(y)}{2}$. But likewise $f(w) = \frac{f(x)+f(z)}{2}$ and $f(w) = \frac{f(y)+f(z)}{2}$. But no two of these quantities can be the same if $f(x)$, $f(y)$, and $f(z)$ are distinct.

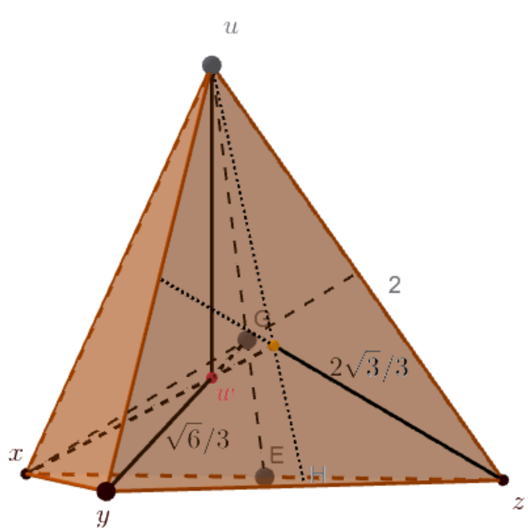
(c) We can imitate what we did in (b), but with a tetrahedron instead. Have node x, y, z, u, and w, set

$$d(x,y) = d(y,z) = d(z,u) = d(u,x) = 2$$
$$d(w,x) = d(w,y) = d(w,z) = d(w,u) = 1.2$$

In a Euclidean space, w must be the midpoint of each edge, but these cannot occur at a single point.



Actually, 1.2 could be replaced by any number h in the interval $[2\sqrt{3}/3, \sqrt{6}/2)$. This is because the distance from the centroid of any face to a vertex is $\frac{2\sqrt{3}}{2}$ while the distance from the centroid of the tetrahedron to any vertex is $\frac{\sqrt{6}}{2}$. The four-element sets form the main tetrahedron and the three *squished* tetrahedron whose apex is w. These can all be realized as long as $h \geq \frac{2\sqrt{3}}{3}$. But putting them altogether would force w to have a distance at least $\frac{\sqrt{6}}{2}$. If $h = \frac{\sqrt{6}}{2}$, then this is realized by the standard tetrahedron.



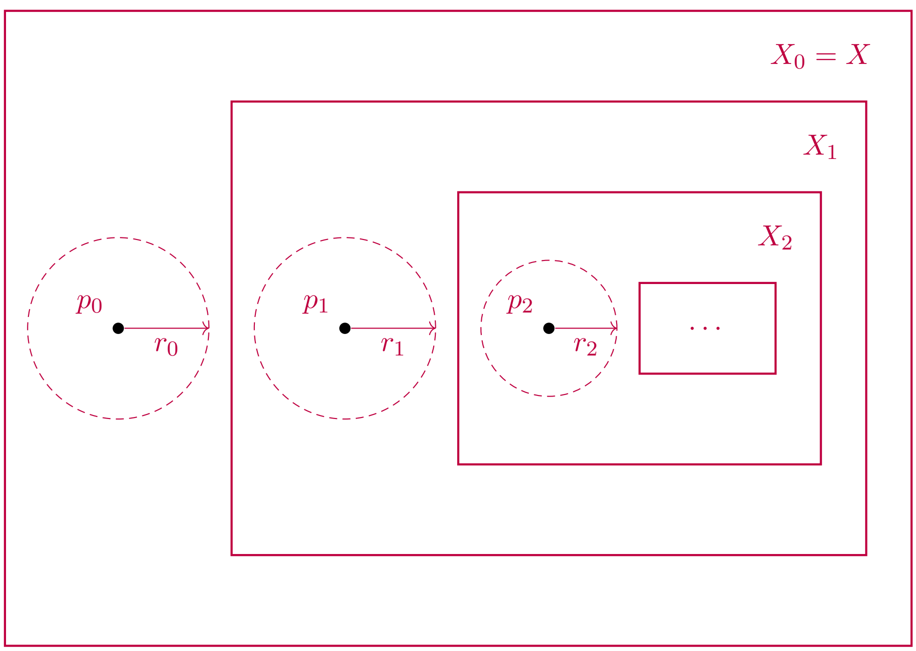
Problem 2.6 (B:2.2:19)**.** Let X be an infinite metric space. Show that there are uncountably many open subsets of X.

(a) Show that you can find $p_i \in X$ and $r_i > 0$ so that $N_{r_i}(p_i) \cap N_{r_j}(p_j) = \emptyset$.

(b) Associate with every $\alpha \in 2^{\mathbb{N}}$ the open set $O_x = \bigcup_{\{i|x(i)=1\}} N_{r_i}(p_i)$. Show that this map is an injection of $2^{\mathbb{N}}$ into the open sets of X.

(c) (b) shows that there are at least continuum many open subsets of X, could the cardinality be even larger?

(a): Take x, y distinct points in X. We can find $r_x, r_y > 0$ so that $N_{r_x}(x) \cap N_{r_y}(y) = \emptyset$ (this is the Hausdorff property of metric spaces) and such that $X \setminus N_{r_x}(x)$ or $X \setminus N_{r_y}(y)$ is infinite (perhaps both are infinite). Suppose $X \setminus N_{r_x}(x)$ is infinite. Then take $p_0 = x$ and $r_0 = r_x$. Now work with $X_1 = X \setminus N_{r_0}(p_0)$. Do the same thing to choose $N_{r_1}(p_1) \subseteq X_1$. So we build $X_0 \supset X_1 \supset X_2 \cdots$ and $p_i \in X_i$ and $r_i > 0$ so that $N_{r_i}(p_i) \subseteq X_i \setminus X_{i+1}$. The following picture makes the idea clear, but in our case $X_{i+1} = X_i \setminus N_{r_i}(p_i)$. The picture looks like we are taking $X_{i+1} \subset X_i \setminus N_{r_i}(p_i)$.



(b): Now for every $\alpha \in 2^{\mathbb{N}}$ define $O_\alpha = \bigcup_{\alpha(i)=1} N_{r_i}(p_i)$. These are open sets, and the map $\alpha \mapsto O_\alpha$ is clearly injective, since if $\alpha \neq \beta$ there is at least i so that $\alpha(i) \neq \beta(i)$. Suppose $\alpha(i) = 1$ and $\beta(i) = 0$, then $N_{r_i}(p_i) \subset O_\alpha$ and $O_\beta \cap N_{r_i}(p_i) = \emptyset$.

(c): There could easily be more than continuum many open sets. Look at the discrete topology on $\mathbb{R}$ with metric $d(x, y) = 1$ if $x \neq y$. Then for any $A \subseteq \mathbb{R}$, $O_A = \bigcup_{x\in A} N_{1/2}(x) = \bigcup_{x\in A}\{x\} = A$ is open. So the cardinality of the open sets is that of $\mathcal{P}(\mathbb{R})$, or $2^{\mathfrak{c}} > \mathfrak{c} = |\mathbb{R}|$.

Problem 2.7 (R:2.26)**.** Let X be a metric space in which every infinite set, A, has a limit point in A. Show that X is compact. (Follow the hints in Rudin.)

This is a very important property. We will see that for metric spaces, this gives the equivalence of "X compact" with "X is sequentially compact."

Following Rudin. First, we show that there is a set $\mathcal{G} = \{G_i \mid i \in \mathbb{N}\}$ so that for all $x \in X$ open set O, there is a G_i such that $x \in G_i \subseteq O$, in other words, X ***has a countable base***.

Step 1: Show that X is ***separable***, that is, X contains a countable dense subset. Fix $\delta > 0$ and $x_0 \in X$, if possible find x_1 so that $d(x_0, x_1) \geq \delta$, continue finding $\{x_0, x_1, \ldots x_n\}$ so that $d(x_i, x_j) \geq \delta$ for $i \neq j$. This process must stop, as otherwise we get an infinite subset $\{x_i \mid i \in \mathbb{N}\}$ with no limit point. So we build for each n a finite set F_n so that for all $x \in X$, there is $y \in F_n$ so that $d(x, y) < 1/n$. Let $F = \bigcup F_n$. Clearly, F is dense and countable.

Step 2: Any separable metric space has a countable base. Let $F = \{p_i \mid i \in \mathbb{N}\}$ be a separable dense set and take as elements of our base $\mathcal{G}$, the sets $N_{1/n}(p_i)$ for all $n, i \in \mathbb{N}$.

Let O be open and $N_r(x) \subseteq O$. Let $p \in F$ be such that $d(p, x) < r/4$ and $r/4 < 1/n < r/2$, here we assume $r < 1$ so that $(2/r, 4/r)$ contains an integer, but we can always make r smaller.

So $x \in N_{1/n}(p)$ and for $y \in N_{1/n}(p)$, $d(x, y) \leq d(y, p) + d(p, x) < 1/n + r/4 < r/2 + r/4 < r$. So $x \in N_{1/n}(p) \subseteq N_r(x) \subseteq O$. So we have a base.

The above captures what we needed from R:2.23 and R:2.24. Now let $\mathcal{O}$ be an open cover of X. Let $\mathcal{O}$ be an open cover of X, consider, $\mathcal{G}' = \{G_i \mid G_i \subseteq O \text{ for some } O \in \mathcal{O}\}$ a subset of our base. $\mathcal{G}'$ is a cover of X since for $x \in X$, $x \in O$ for some $O \in \mathcal{O}$, but then $x \in G_i \subseteq O$ for some i. Now for each $G_i \in \mathcal{G}'$ choose one $O \in \mathcal{O}$ so that $G_i \subseteq O$. Let $\mathcal{O}'$ be the set of all selected O. Since $\mathcal{G}'$ is a cover, so is $\mathcal{O}'$, and clearly $\mathcal{O}'$ is countable. So let $\mathcal{O}' = \{O_i \mid i \in \mathbb{N}\}$.

Suppose no finite subset covers X and consider the increasing sequence $V_i = \bigcup_{j \leq i} O_i$ of open sets. By dropping some sets if necessary, we may assume that the V_i's are strictly increasing. Let $C_i = X \setminus V_i$ so C_i is a strictly decreasing sequence of closed sets with empty intersection since $\bigcup_{i \in \mathbb{N}} V_i = X$. Let $x_i \in C_i \setminus C_{i+1}$. The set of points $P = \{x_i \mid i \in \mathbb{N}\}$ is an infinite set and by assumption has a limit point x. But $x \in C_i$ since clearly x is a limit point of $C_i \cap P$ for all i and C_i is closed. This contradicts the fact that $\bigcap_{i \in \mathbb{N}} C_i = \emptyset$.

Problem 2.8 (B:2.3:3 - Compact sets get crowded!)**.** Let X be a compact metric space and let $\epsilon > 0$. Show that there is an N so that for any $m \geq N$, if $p_1, \ldots, p_m$ are m distinct points in X, then at least two of them are within ϵ of each other, that is, there are i, j with $d(p_i, p_j) < \epsilon$.

Let $\mathcal{O} = \{N_{\epsilon/2}(x) \mid x \in X\}$. This is an open cover of X s by compactness, there is a finite subcover, $\mathcal{O}' = \{N_{\epsilon/2}(x_1), \ldots, N_{\epsilon/2}(x_k)\}$. Let $m > k$ and take $\{p_1, \ldots p_m\}$ to be m distinct points of X. By the pigeonhole property, there are distinct i, j so that $p_i, p_j \in N_{\epsilon/2}(x_t)$ for some t. But then $d(p_i, p_j) \leq d(p_i, x_t) + d(x_t, p_j) < \epsilon/2 + \epsilon/2 = \epsilon$.

Problem 2.9 (B:2.5:1 - The usual definition of connected.)**.** (a) Show that $E \subsetneq X$ is connected (as per Rudin) iff there is no non-empty proper subset of E that is ***clopen*** relative to E, i.e., closed and open relative to E.

(b) Show that $E \subsetneq X$ is connected (as per Rudin) iff there are two non-empty ***relatively open subsets*** of E that ***cover*** E.

Note: A space X is connected iff the clopen sets consist of exactly X and $\emptyset$.

Rudin's definition is: $E \subseteq X$ is *connected* iff E is not the union of two non-empty separated sets, where A, B are *separated* iff $A \cap \mathrm{Cl}(B) = \emptyset = \mathrm{Cl}(A) \cap B$.

Suppose $A \neq \emptyset$ is clopen relative to E. So $A = E \cap U = E \cap V$ and $B = E \setminus A = E \cap U^c = E \cap V^c$ for some open U and closed V. So $\mathrm{Cl}(B) = \mathrm{Cl}(E \cap U^c) \subseteq \mathrm{Cl}(U^c) = U^c$ since U^c is closed. But then clearly $A \cap \mathrm{Cl}(B) = \emptyset$. Similarly, $\mathrm{Cl}(A) = \mathrm{Cl}(E \cap V) \subseteq \mathrm{Cl}(V) = V$ and so $\mathrm{Cl}(A) \cap B = \emptyset$ since $B \subseteq V^c$. Thus, we see that A and B are separated and thus E fails to be Rudin connected.

Conversely, if $E = A \cup B$ where A and B are non-empty and separated, then $A = \mathrm{Cl}(A) \cap E = \mathrm{Cl}(B)^c \cap E$ and hence A clopen relative to E.

So (a) is done. (b) follows easily since if $A \subsetneq E$ is a clopen relative to E non-empty proper subset of E, then this is also true of $B = E \setminus A$. Conversely, if $A \cup B = E$ and A, B are open relative to E, then $E \setminus A = B$ is closed relative to E, and similarly A is closed relative to E, so A and B are non-empty proper clopen subsets of E.

Problem 2.10 (This is a mix of problems.)**.** Consider the following collections of open subsets of $\mathbb{Q}$. (1) Verify that each gives a topology on $\mathbb{Q}$ and (2) compute for each, $\mathrm{Cl}((0, \sqrt{2}))$, (3) show that the space is compact, or not, (4) show that the space is connected or not, and finally (5) how are these related to the standard relative topology that $\mathbb{Q}$ inherits from $\mathbb{R}$?

(a) *The right-ray topology.* Take $\mathcal{O}$ to be the set of all right-rays of the form (a, ∞) for $a \in \mathbb{R} \cup \{+\infty, -\infty\}$.

(b) *The co-finite topology.* Take $\mathcal{O}$ to be subsets of $\mathbb{Q}$ whose complements are finite together with $\emptyset$.

(c) *The order topology.* A set A is open in this space iff for all $q \in A$, there is $q_1 < q < q_2$ in $\mathbb{Q}$ so that $(q_1, q_2) \cap Q \subseteq A$.

Right-ray topology: (1) This has $\mathbb{Q}$ and $\emptyset$ by definition. Also, arbitrary unions and intersections of these rays are again a ray. So this is a topology.

(2) Take any $x < \sqrt{2}$ let (a, ∞) be a ray containing x, then clearly $(a, \infty) \cap (0, \sqrt{2}) \neq \emptyset$. So $x \in \mathrm{Cl}((0, \sqrt{2})$. Hence $\mathrm{Cl}((0, \sqrt{2}) = (-\infty, \sqrt{2})$.

(3) This is not compact since, for example, there is no finite subcover of $\{(-n, \infty) \mid n \in \mathbb{N}\}$.

(4) There are no disjoint open subsets, so the topology is connected.

(5) Every open set here is open in the standard topology on $\mathbb{Q}$ inherited from $\mathbb{R}$.

Co-finite topology (1) Clearly $\mathbb{Q}$ and $\emptyset$ are open. The union of sets will have a smaller complement than any of its elements, so these are closed under union. A finite intersection of co-finite sets is still co-finite. So we have a topology.

(2) The only closed sets are finite sets or the entire space. So $\mathrm{Cl}((0, \sqrt{2}) = \mathbb{Q}$.

(3) The topology is compact. Given an open cover, fix a single element O. There are only finitely many points to cover, so we easily extend to a finite subcover.

(4) There are no non-empty disjoint proper open subsets.

(5) Again, the open sets here are open in the standard topology on $\mathbb{Q}$ inherited from $\mathbb{R}$.

The order topology: (1) The open sets of the order topology consist of unions of open intervals. This is exactly the same as the topology inherited from $\mathbb{R}$ using the standard metric.

(2) $\mathrm{Cl}(0, \sqrt{2}) = [0, \sqrt{2})$.

(3) Non-compact, consider $\mathcal{O} = \{(-n, n) \mid n \in \mathbb{N}\}$.

(4) Not connected, for example, $A = (-\infty, \sqrt{2})$ and $B = (\sqrt{2}, \infty)$ exemplify this. In fact (α, β) for any irrationals $\alpha < \beta$ is clopen. So $\mathbb{Q}$ has a clopen base! This is called ***totally disconnected***.

(5) As mentioned in (1), this is exactly the same as the inherited topology from $\mathbb{R}$.